

Weekly training log, week 10

Week 10

Module	EEEM068, Applied Machine Learning
Project	Industrial waste classification on WaRP
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Focus	Faster R-CNN on WaRP-D

1. Experimental settings

Dataset	WaRP-D, 2452 train, 522 val, 28 classes
GPU	NVIDIA RTX 4000 Ada, 20 GB
Metric	COCO mAP at IoU 0.5 and mAP from 0.5 to 0.95
Goal	Two stage detector baseline on WaRP-D.

2. Hyperparameters

Trial 1

Backbone	Faster R-CNN with ResNet-50 FPN, COCO pretrained
Batch size	4
Epochs	15 (5 heads only, then 10 full)
Phase 1 LR	1e-4
Phase 2 LR	head 5e-5, backbone 5e-6
Weight decay	1e-4
Optimizer	AdamW
Loss	standard Faster R-CNN loss (RPN + box + cls)
Params	41.2 million

3. Results

Trial 1, Phase 1 (heads only)

Epoch	Train loss	Val loss
1	0.787	0.630
2	0.669	0.596
3	0.615	0.563
4	0.579	0.551
5	0.547	0.537

Val loss down from 0.79 to 0.54 across the 5 head only epochs.

Trial 1, Phase 2 (full fine tuning)

Epoch	Train	Val loss	mAP@50	mAP 50-95
1	0.546	0.454	40.05%	25.15%
3	0.403	0.414	52.53%	34.93%
5	0.312	0.415	54.44%	36.44%
7	0.232	0.413	55.82%	38.59%
9	0.189	0.430	56.43%	39.80%
10	0.177	0.432	56.21%	39.66%

Best mAP@50: 56.43%. Val loss starts creeping up after epoch 6, mild overfitting but mAP keeps going up.

4. Observations

The big jump happens at epoch 1 of Phase 2, when the backbone unfreezes. The model went from zero to 40% mAP@50 in one epoch.

15 epochs total was enough. Going longer would just overfit on 2452 images.

Final result: 56.43% mAP@50 reported in the table.